

A Philosophical-Mathematical Representation of Deductive and Inductive Logic: A Diagrammatic and Formal Analysis

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Abstract

We give a diagrammatic representation of deductive and inductive reasoning in which deduction is a certain passage from thought to existence, $\varphi \rightarrow \exists$, and induction is a tentative, iterated passage from the study of existence to the theory of thought, $s\exists \circ t\varphi$. We make the diagram precise in a possible-worlds semantics: thoughts are propositions about worlds, deduction is semantic entailment, polarity $\uparrow, \downarrow$ distinguishes affirmation from modus tollens, and an inductive theory is supported by a study when some world is consistent with both. Within this setting we prove that deduction is sound and transitive, that iterated observation can only narrow the supported theories, that a single counterexample falsifies a universal theory, and that no finite study entails a universal theory over an unobserved individual. The last statement is a formal version of Hume’s problem of induction. As an application we realize the framework in quantum information theory: thoughts become subspaces, deduction becomes certain measurement and unitary evolution, induction becomes state tomography, and the classical setting is recovered as the case of commuting observables. The quantum model reproduces refinement, falsification and underdetermination, and adds a limitation with no classical counterpart: measurements in a single basis never determine a quantum state. Every theorem in Sections 4–6, and the three propositions of the quantum model, is verified in the Lean 4 proof assistant; Appendix A lists the corresponding declarations.

Contents

1	Introduction	2
2	The diagrammatic framework	3
3	Formal setting	4
4	Deduction	4
5	Induction	5
5.1	Iterative refinement	5
5.2	Falsification and the problem of induction	5
6	The foundation	6

7	Application: a quantum information model	6
7.1	Dictionary	6
7.2	Deduction as certain measurement	7
7.3	Induction as state tomography	8
7.4	Foundation: qubits and spins	8
7.5	Physical implementations	9
7.6	Thermodynamic reading	9
7.7	Experimental proposals	9
8	Discussion	10
9	Conclusion	11
A	Lean formalization	11

1 Introduction

Deductive logic derives particular conclusions from general premises with certainty: all men are mortal; Socrates is a man; therefore Socrates is mortal [1]. Inductive logic builds general theories from particular observations with only tentative support: many swans have been seen to be white, so perhaps all swans are white. Textual arguments and static diagrams, such as Venn diagrams and Peirce’s existential graphs [16], show the structure of a single inference well but do not show reasoning as a process in which deduction and induction alternate.

This paper proposes a diagram (Figure 1) that places both modes of reasoning in one picture, with a straight arrow for deduction, a circular arrow for induction, and a foundation of iterations, hypotheses, levels, binary states and polarities. We then give the diagram a precise semantics and prove what it asserts.

Contributions.

1. A diagrammatic framework for deduction $\varphi \rightarrow \exists$ and induction $s\exists \circ t\varphi$ (Section 2).
2. A possible-worlds semantics for every symbol of the diagram (Section 3).
3. Theorems for each mode: soundness, transitivity and modus tollens for deduction (Section 4); monotone refinement, falsification and the failure of induction as deduction (Theorem 5.4) for induction (Section 5); and a truth-value characterization of the foundation (Section 6).
4. An application to quantum information theory (Section 7): a dictionary from the framework to quantum states, measurements and Hamiltonians; quantum counterparts of the deductive and inductive theorems (Propositions 7.1 and 7.3); the classical framework as the commuting case (Proposition 7.2); and physical implementations, a thermodynamic reading and experimental proposals.
5. A machine-checked formalization of the results of Sections 4–6 in Lean 4 [6], and of Propositions 7.1 to 7.3 with Mathlib [19] (Appendix A).

2 The diagrammatic framework

The diagram has two primary rows and a foundation (Figure 1).

- **Deduction**, $\varphi \rightarrow \exists$. A thought φ (premises) leads to an existence $\exists$ (a conclusion) along a straight arrow, read *hence*. The passage is certain. Polarity $\uparrow$ marks an affirmed conclusion and $\downarrow$ a negated one.
- **Induction**, $s\exists \circlearrowright t\varphi$. The study of existence $s\exists$ (observations) leads to a theory of thought $t\varphi$ (a hypothesis) along a circular arrow, read *conclude*. The passage is tentative and is repeated as new observations arrive.
- **Foundation**. Iterations i , hypotheses h and levels l , together with binary states $0, 1$ and polarities $\uparrow, \downarrow$, generate the thoughts used above.

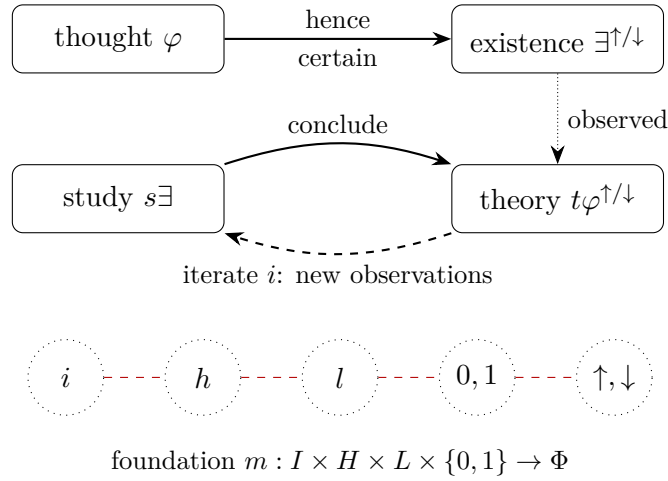


Figure 1: Deduction (top row), induction (middle row) and the foundation (bottom).

Table 1 fixes the reading of each symbol. One change from the 2025 version: the symbol $\exists$ is kept for the diagram only, and the set of possible existences is written $\mathcal{W}$, so that it is not confused with the existential quantifier.

Symbol	Informal meaning	Formal reading (Section 3)
φ	thought	proposition $\varphi : \mathcal{W} \rightarrow \text{Prop}$
$\exists$	existence	proposition about worlds; $\mathcal{W}$ is the set of worlds
s	study	list of observations
t	theory	proposition about worlds
$\rightarrow$	hence	semantic entailment $\models$
$\circlearrowright$	conclude	support (Definition 3.4)
i	iteration	adding an observation to a study
h, l	hypothesis, level	indices of the foundation map m
$0, 1$	binary state	truth values
$\uparrow, \downarrow$	polarity	affirmation, negation

Table 1: Symbols of the framework.

3 Formal setting

Fix a set $\mathcal{W}$ of *possible worlds*. Every symbol of the diagram is read as follows.

Definition 3.1 (Thought). A *thought* is a proposition about worlds, that is, a map $\varphi : \mathcal{W} \rightarrow \text{Prop}$. We write Φ for the set of thoughts and $\varphi(w)$ for “ φ holds in w ”. An existence is likewise a thought; the distinction between φ and $\exists$ in the diagram is one of role (premise versus conclusion), not of type.

Definition 3.2 (Entailment). A thought φ *entails* ψ , written $\varphi \models \psi$, if $\{w \in \mathcal{W} : \varphi(w)\} \subseteq \{w \in \mathcal{W} : \psi(w)\}$.

Entailment is a relation, not a function: a thought entails many conclusions. This replaces the map $f : \Phi \rightarrow \exists$ of the 2025 version, which could not assign a single conclusion to each thought.

Definition 3.3 (Polarity). The polarities are $\uparrow$ and $\downarrow$. For a proposition p , set $p^\uparrow = p$ and $p^\downarrow = \neg p$.

Definition 3.4 (Study and support). A *study* is a finite list $s = (o_1, \dots, o_n)$ of thoughts, the observations. A theory t is *supported* by s , written $s \circ t$, if some world satisfies t and every o_k :

$$s \circ t \quad :\Longleftrightarrow \quad \exists w \in \mathcal{W}, t(w) \wedge \bigwedge_{k=1}^n o_k(w).$$

Remark 3.5. The 2025 version required $P(T_\varphi \mid S_\exists) > 0$. That quantity is undefined when the study has probability zero. If $\mathcal{W}$ is countable and P gives every world positive probability, then $P(t \wedge s) > 0$ holds exactly when $s \circ t$, so [Definition 3.4](#) is the qualitative content of the original condition and needs no choice of prior. This remark is not part of the Lean formalization.

4 Deduction

Proposition 4.1 (Material implication). *For propositions p, q , $(p \rightarrow q) \iff (\neg p \vee q)$.*
imp_iff_not_or

Proof. If $p \rightarrow q$, then either p holds and so q holds, or $\neg p$ holds; this case split uses the law of excluded middle. Conversely, if $\neg p \vee q$ and p , the first disjunct is impossible, so q . $\square$

Theorem 4.2 (Deduction is certain). *If $\varphi \models \psi$ and $\varphi(w)$, then $\psi(w)$.* *deduction_sound*

Proof. By [Definition 3.2](#), w lies in the set of worlds satisfying φ , which is contained in the set satisfying ψ . $\square$

Proposition 4.3 (Chains of deduction). *If $\varphi \models \psi$ and $\psi \models \chi$, then $\varphi \models \chi$.* *Entails.trans*

Proof. Set inclusion is transitive. $\square$

Proposition 4.4 (Negative polarity is modus tollens). *If $\varphi \models \psi$ and $\psi(w)^\downarrow$, then $\varphi(w)^\downarrow$.*
modus_tollens

Proof. If $\varphi(w)$ held, [Theorem 4.2](#) would give $\psi(w)$, contradicting $\neg\psi(w)$. $\square$

Example 4.5 (Deductive swan). Let B be a set of birds with predicates Swan and White. If $\forall b, \text{Swan}(b) \rightarrow \text{White}(b)$ and $\text{Swan}(a)$, then $\text{White}(a)$. In the diagram this is $\varphi \rightarrow \exists^\uparrow$. *swan_syllogism*

5 Induction

5.1 Iterative refinement

Iteration i adds an observation to a study. The following results say that iteration can only narrow the set of supported theories, which is the precise sense in which induction refines.

Proposition 5.1. *If $(o, o_1, \dots, o_n) \circ t$, then $(o_1, \dots, o_n) \circ t$.* *Supported.of_cons*

Theorem 5.2 (Refinement is monotone). *If every observation of s occurs in s' and $s' \circ t$, then $s \circ t$.* *Supported.mono*

Proof. A world witnessing $s' \circ t$ satisfies t and every observation of s' , hence every observation of s . [Proposition 5.1](#) is the case $s' = (o) \dashv\vdash s$. □

5.2 Falsification and the problem of induction

For the swan examples take worlds to be colourings $c : B \rightarrow \{\text{true}, \text{false}\}$ recording which birds are white. For a finite list *seen* of birds define

$$\text{SeenWhite}_{\text{seen}}(c) :\iff \forall x \in \text{seen}, c(x) = \text{true}, \quad \text{AllWhite}(c) :\iff \forall x \in B, c(x) = \text{true}.$$

Proposition 5.3 (Falsification). *If $\text{Swan}(b)$ and $\neg \text{White}(b)$, then $\neg \forall x, \text{Swan}(x) \rightarrow \text{White}(x)$.*
falsification

Proof. Instantiate the universal statement at b . □

Falsification is deductive: a single observation settles the matter, in line with Popper [17]. Confirmation is not, however many observations are made.

Theorem 5.4 (Induction is not deduction). *Let B have decidable equality, let *seen* be a finite list of birds, and let $b \in B$ with $b \notin \text{seen}$. Then $\text{SeenWhite}_{\text{seen}} \not\models \text{AllWhite}$.* *induction_not_deduction*

Proof. Define $c(x) = \text{false}$ if $x = b$ and $c(x) = \text{true}$ otherwise. Every $x \in \text{seen}$ differs from b , so $c(x) = \text{true}$ and $\text{SeenWhite}_{\text{seen}}(c)$ holds. But $c(b) = \text{false}$, so $\text{AllWhite}(c)$ fails. Hence the worlds satisfying $\text{SeenWhite}_{\text{seen}}$ are not contained in those satisfying AllWhite . □

Proposition 5.5 (Inductive swan). *For every finite list *seen*, $(\text{SeenWhite}_{\text{seen}}) \circ \text{AllWhite}$.*
allWhite_supported

Proof. The colouring that makes every bird white satisfies both. □

Together, [Theorem 5.4](#) and [Proposition 5.5](#) state the asymmetry the diagram depicts: the theory “all swans are white” remains supported by every finite study, so it may be held along the circular arrow, but no finite study entails it, so it never travels the straight arrow. This is a formal version of Hume’s problem of induction [9].

6 The foundation

Definition 6.1 (Foundation map). Let I, H, L be sets of iterations, hypotheses and levels, and let $\beta : I \times H \times L \rightarrow \Phi$ assign a base thought to each index. The *foundation map* $m : I \times H \times L \times \{0, 1\} \rightarrow \Phi$ is

$$m(i, h, l, 1) = \beta(i, h, l), \quad m(i, h, l, 0) = \neg\beta(i, h, l).$$

Thus state 1 carries polarity $\uparrow$ and state 0 carries polarity $\downarrow$. In the 2025 version m took values in $\Phi \cup \exists$; since existences are thoughts (Section 3), the codomain is simply Φ .

Proposition 6.2 (The binary state is the truth value). *For every world w and $b \in \{0, 1\}$, $m(i, h, l, b)(w)$ holds if and only if $\beta(i, h, l)(w)$ agrees with b , that is, $\beta(i, h, l)(w) \iff b = 1$.*
 m_spec

Proof. For $b = 1$ both sides say $\beta(i, h, l)(w)$. For $b = 0$ both sides say $\neg\beta(i, h, l)(w)$. $\square$

7 Application: a quantum information model

This section gives a physical realization of the framework in quantum information theory. Thoughts become subspaces of a Hilbert space, deduction becomes certain measurement and unitary evolution, induction becomes quantum state tomography, and the binary and polarized foundation becomes qubits and spins. The classical setting of Section 3 reappears as the special case in which all observables commute (Proposition 7.2). Propositions 7.1–7.3 are formalized in Lean with Mathlib (Appendix A); the surrounding discussion of unitary evolution, physical implementations and thermodynamics is not.

Throughout, $\mathcal{H}$ is a finite-dimensional complex Hilbert space, a *state* is a unit vector $|v\rangle \in \mathcal{H}$ or a density operator ρ , and a *projector* is an operator P with $P = P^\dagger = P^2$. For projectors, $P \leq Q$ means $\text{ran } P \subseteq \text{ran } Q$, equivalently $QP = P$. Physical time is written τ , to keep t for theories.

7.1 Dictionary

Symbol	Quantum entity	Physical meaning
φ (thought)	projector P_φ , or a state $ v\rangle$	a property; a superposition of possibilities
$\exists$ (existence)	eigenstate $ n\rangle$	a definite measurement outcome
$\models$ (hence)	$P_\varphi \leq P_\psi$	subspace inclusion
s (study)	outcomes of measurements $\{P_n\}$	observed statistics
t (theory)	reconstructed state $\hat{\rho}$, or model $U(\theta)$	hypothesis about the system
i (iteration)	$U(\tau) = e^{-i\hat{H}\tau/\hbar}$, circuit layers, adaptive rounds	refinement in time
h (hypothesis)	measurement basis $\{ n\rangle\}$	what is asked of the system
l (level)	energy levels E_n of $\hat{H}$	hierarchical structure
$0, 1$ (binary)	qubit basis $ 0\rangle, 1\rangle$	computational basis
$\uparrow, \downarrow$ (polarity)	spin states $ \uparrow\rangle, \downarrow\rangle$; $P^\downarrow = I - P$	affirmation, negation

Table 2: Quantum realization of the symbols of Table 1.

7.2 Deduction as certain measurement

A thought φ is realized by a projector P_φ : the property “the system lies in $\text{ran } P_\varphi$ ”. Entailment is subspace inclusion, and negative polarity is the orthogonal complement, $P^\perp = I - P$.

Proposition 7.1 (Quantum deductive certainty). *Let $P_\varphi \leq P_\psi$ and let $|v\rangle$ be a state with $P_\varphi|v\rangle = |v\rangle$. Then measuring $\{P_\psi, I - P_\psi\}$ on $|v\rangle$ yields P_ψ with probability $\langle v|P_\psi|v\rangle = 1$. Moreover $I - P_\psi \leq I - P_\varphi$.*
quantum_deduction_sound

Proof. Since $P_\psi P_\varphi = P_\varphi$, we get $P_\psi|v\rangle = P_\psi P_\varphi|v\rangle = P_\varphi|v\rangle = |v\rangle$, so $\langle v|P_\psi|v\rangle = \langle v|v\rangle = 1$. For the second claim, $(I - P_\varphi)(I - P_\psi) = I - P_\psi - P_\varphi + P_\varphi P_\psi = I - P_\psi$, using $P_\varphi P_\psi = (P_\psi P_\varphi)^\dagger = P_\varphi$. $\square$

Proposition 7.1 is the quantum form of Theorem 4.2 and Proposition 4.4. Unitary evolution is deductive in the same sense: $|v\rangle \mapsto U|v\rangle$ is deterministic, and if $UP_\varphi U^\dagger \leq P_\psi$ then every state with property φ is carried to one with property ψ . A general measurement is not certain. For a projective measurement $\{P_n\}$ with $\sum_n P_n = I$ and $P_n P_m = \delta_{nm} P_n$, the Born rule gives

$$|v\rangle \mapsto \frac{P_n|v\rangle}{\|P_n|v\rangle\|} \quad \text{with probability } p(n) = \langle v|P_n|v\rangle.$$

What is deduced from the state is the distribution p , not the individual outcome. For a spin, $|v\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ yields $|\uparrow\rangle$ with probability $|\alpha|^2$ and $|\downarrow\rangle$ with probability $|\beta|^2$; certainty is the case $|\alpha| = 1$ or $|\beta| = 1$.

Proposition 7.2 (The classical framework is the commuting case). *Let $\{|n\rangle\}_{n \in \mathcal{W}}$ be an orthonormal basis indexed by the finite set of worlds $\mathcal{W}$, and set $P_\varphi = \sum_{n: \varphi(n)} |n\rangle\langle n|$. Then*

1. $P_\varphi \leq P_\psi$ if and only if $\varphi \models \psi$, and $P_{\neg\varphi} = I - P_\varphi$; *classicalProj_le_iff, classicalProj_not*
2. if $\rho = \sum_n p_n |n\rangle\langle n|$ with every $p_n > 0$, then for a study $s = (o_1, \dots, o_k)$ and theory t ,
 $\text{Tr}(\rho P_t P_{o_1} \cdots P_{o_k}) > 0$ if and only if $s \circlearrowleft t$. *classical_trace_pos_iff*

Proof. (1) $\text{ran } P_\varphi$ is spanned by the $|n\rangle$ with $\varphi(n)$, and such a basis vector lies in $\text{ran } P_\psi$ exactly when $\psi(n)$. The second claim holds because every basis vector satisfies exactly one of φ and $\neg\varphi$. (2) The projectors are diagonal, so their product is the projector onto the $|n\rangle$ with $t(n) \wedge o_1(n) \wedge \cdots \wedge o_k(n)$, and the trace is $\sum p_n$ over those n . A sum of positive numbers is positive exactly when it is non-empty, which is Definition 3.4. $\square$

Part (2) realizes Remark 3.5 physically. Outside the commuting case the projectors of $\mathcal{H}$ do not form a Boolean algebra [5, 20]. For a spin, with P_{x+} the projector onto $(|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$,

$$P_{x+} \wedge (P_{z+} \vee P_{z-}) = P_{x+} \wedge I = P_{x+}, \quad (P_{x+} \wedge P_{z+}) \vee (P_{x+} \wedge P_{z-}) = 0,$$

so the distributive law fails. This marks where the quantum model goes beyond Section 4: the identity of Proposition 4.1 relies on a Boolean structure that incompatible observables lack.

7.3 Induction as state tomography

In the quantum model a study is a record of outcomes $n_1, \dots, n_N$ obtained by measuring identical copies of an unknown state, and a theory is a state ρ that explains them. The likelihood of ρ is $L_N(\rho) = \prod_{k=1}^N \text{Tr}(\rho P_{n_k})$, and ρ is *supported* when $L_N(\rho) > 0$. Maximum-likelihood tomography selects

$$\hat{\rho}_{\text{ML}} = \arg \max_{\rho} \prod_{k=1}^N \text{Tr}(\rho P_{n_k}),$$

and adaptive tomography chooses the next basis from the outcomes so far,

$$h_{k+1} = f(n_1, \dots, n_k; h_1, \dots, h_k).$$

This is the circular arrow $s\exists \circ t\varphi$ with iteration i indexing the rounds.

Proposition 7.3 (Quantum refinement, falsification and underdetermination).

1. Each further outcome can only shrink the set of supported states: $L_{N+1}(\rho) > 0$ implies $L_N(\rho) > 0$. *QSupported.of_concat*
2. Suppose N measurements of $\{|\uparrow\rangle\langle\uparrow|, |\downarrow\rangle\langle\downarrow|\}$ all give $\uparrow$. Then $|\uparrow\rangle$ is the maximum-likelihood state, yet every state $|\theta, \phi\rangle = \cos\theta|\uparrow\rangle + e^{i\phi}\sin\theta|\downarrow\rangle$ with $0 \leq \theta < \pi/2$ is supported, with likelihood $\cos^{2N}\theta > 0$. *spinUp_maxLikelihood, spinState_supported*
3. A single outcome $\downarrow$ refutes the theory $|\uparrow\rangle$: its likelihood becomes 0. *spinUp_refuted*
4. No number of measurements in this basis distinguishes $|\theta, \phi\rangle$ from $|\theta, \phi'\rangle$: their outcome statistics are equal for all ϕ, ϕ' . *likelihood_spinState_phase*

Proof. (1) $L_{N+1}(\rho) = L_N(\rho) \text{Tr}(\rho P_{n_{N+1}})$, and every factor $\text{Tr}(\rho P)$ lies in $[0, 1]$, so $0 \leq L_{N+1}(\rho) \leq L_N(\rho)$. (2) $\langle\theta, \phi|\uparrow\rangle\langle\uparrow|\theta, \phi\rangle = \cos^2\theta$, so the likelihood is $\cos^{2N}\theta$, which is positive for $\theta < \pi/2$ and largest at $\theta = 0$. (3) $\langle\uparrow|\downarrow\rangle\langle\downarrow|\uparrow\rangle = 0$. (4) The outcome probabilities are $\cos^2\theta$ and $\sin^2\theta$, independent of ϕ . $\square$

Parts (1)–(3) are the quantum counterparts of [Theorem 5.2](#), [Theorem 5.4](#) and [Proposition 5.3](#). Part (4) is a limitation with no classical counterpart: data from one basis leave the phase undetermined however much of it is collected, so a tomographically complete set of bases is needed. In the language of the framework, the hypothesis h (the basis) bounds what any study can reveal.

7.4 Foundation: qubits and spins

The foundation map m of [Section 6](#) is realized by the projector onto $|b\rangle$ in the basis selected by h : state 1 affirms and state 0 negates, as in [Proposition 6.2](#). A register of n qubits with spin degrees of freedom has states

$$|v\rangle = \sum_{x \in \{0,1\}^n} \sum_{\sigma \in \{\uparrow, \downarrow\}^n} c_{x,\sigma} |x\rangle |\sigma\rangle,$$

and iteration is a circuit of depth d , $|v_{\text{final}}\rangle = U_d \cdots U_2 U_1 |v_{\text{initial}}\rangle$.

7.5 Physical implementations

Deduction: quantum gates. Single-qubit gates (X , Y , Z , Hadamard) and two-qubit gates (CNOT, CZ) act as $|v\rangle \mapsto U|v\rangle$, deterministic transformations that realize the straight arrow [15, 8].

Induction: variational algorithms. The variational quantum eigensolver and the quantum approximate optimization algorithm prepare $|v(\theta)\rangle = U(\theta)|0\rangle$ and update

$$\theta_{k+1} = \theta_k - \eta \nabla_{\theta} \langle v(\theta_k) | \hat{H} | v(\theta_k) \rangle,$$

with the gradient estimated from measurement outcomes. The parameters are the theory, the measurements are the study, and each update is one turn of the circular arrow [18].

Induction: quantum machine learning. Training data $\{(x_k, y_k)\}$ form the study, a parametrized circuit $U(\theta)$ is the theory, and training minimizes $L(\theta) = \sum_k |y_k - \langle v(x_k) | M | v(x_k) \rangle|^2$ for an observable M [4].

7.6 Thermodynamic reading

Information has a physical cost. Recording a study of N binary outcomes uses N bits, and by Landauer's principle resetting that record for reuse dissipates at least $Nk_B T \ln 2$ of heat [12, 3]. The two arrows then differ thermodynamically:

- *Deduction preserves entropy.* Unitary evolution leaves the von Neumann entropy $S(\rho) = -\text{Tr } \rho \ln \rho$ unchanged, $S(U\rho U^\dagger) = S(\rho)$; no information is gained or lost.
- *Induction lowers the observer's entropy, at a price.* Observation reduces the expected Shannon entropy of the observer's belief about the theory, $H(T | S) \leq H(T)$. As with Maxwell's demon, the reduction is paid for by the Landauer cost of erasing the record [3].

7.7 Experimental proposals

1. *Tomography.* Prepare an unknown state $|v\rangle$, measure in a tomographically complete set of bases, reconstruct $\hat{\rho}$ by maximum likelihood, and report the fidelity $F = \langle v | \hat{\rho} | v \rangle$. Plotting F against N shows inductive refinement directly.
2. *Quantum walk.* The state $|v(\tau)\rangle = \sum_n \alpha_n(\tau) |n\rangle |c_n\rangle$, with position $|n\rangle$ and coin $|c_n\rangle$, displays iteration as coherent evolution.
3. *Error correction.* A logical qubit $|v_L\rangle$ is a protected thought, the noisy physical qubits are existence, the syndrome measurement is the study $s\exists$, and the decoder's inferred error is the theory $t\varphi$; applying the correction is then a deduction.

8 Discussion

Philosophy. The straight arrow carries Descartes’ movement from thought to existence [7]: what is entailed holds in every world where the premises hold (Theorem 4.2). The circular arrow carries Hume’s and Popper’s view of induction: a generalization is supported, never entailed (Theorem 5.4), and can be overturned by a single counterexample (Proposition 5.3). Levels l in the foundation give a place for Kant’s hierarchical structuring of thought [11].

Quantum analogies. The quantum model of Section 7 suggests three further analogies, which we state as analogies rather than results. *Complementarity*: as quantum objects show wave and particle aspects, reasoning involves both abstract thought (φ) and concrete existence ($\exists$). *Uncertainty*: incompatible observables cannot both be sharp, which is a physical limitation on what any study can reveal, distinct from Hume’s logical one (Proposition 7.3(4)). *Entanglement*: entangled systems show correlations that no classical model reproduces, suggesting that logical relations between parts of a system need not reduce to relations between the parts; these correlations cannot be used to signal, so they do not conflict with causality. These views connect to the idea that physical processes are computations [8, 14].

Education and cognition. The top row of the diagram is suitable for a first course in logic; the foundation invites discussion of polarity and iteration. The visual layout follows the view that diagrams make inferential structure explicit [13, 10].

Limitations and future work.

- Support (Definition 3.4) is qualitative. A quantitative reading of the inductive polarity $t\varphi^\uparrow$ is Bayesian confirmation, $P(t \mid s) > P(t)$; formalizing it requires a probability library such as Mathlib.
- The indices i, h, l of the foundation are uninterpreted sets. Giving levels an order (for example observational below theoretical) would let one state that iteration moves upward.
- Only propositional and first-order statements over a fixed set of worlds are treated. Multi-premise defeasible reasoning and higher-order logic are left open.
- The quantum model is limited in practice by decoherence, state-preparation and measurement errors, and scale [18, 2]. Error-corrected and topological implementations, and hybrid quantum–classical reasoning, are natural next steps.
- The formalization of Section 7 covers its three propositions only. The claims about unitary evolution, the failure of distributivity, and von Neumann entropy are not formalized.
- The diagram’s usefulness for learners is an empirical claim that remains to be tested.

9 Conclusion

We have given a diagram for deduction and induction, a possible-worlds semantics for its symbols, and machine-checked theorems for what it depicts: deduction is certain and transitive, negative polarity is modus tollens, iteration narrows the supported theories, one counterexample falsifies, and no finite study entails a universal theory. Realized in quantum information theory, the same structure appears as certain measurement and unitary evolution on one side and state tomography on the other, with the classical framework as the commuting case and with a new, physical limit on induction. This suggests that quantum mechanics offers a natural physical setting for the interplay of deduction and induction.

A Lean formalization

All results in Sections 4–6 are formalized in Lean 4 (toolchain v4.34.1) in the file `lean/WangLogic/Basic.lean` accompanying this paper, using only the core library. Propositions 7.1 to 7.3 are formalized in `lean/WangLogic/Quantum.lean` with Mathlib [19] (commit d13f23b7). Neither file contains `sorry`, and every declaration depends at most on the axioms `propext`, `Quot.sound` and `Classical.choice`. The project builds with

```
cd lean && lake exe cache get && lake build
```

The file `lean/ComparatorChallenges/InductionNotDeduction.json` states Theorem 5.4 for independent checking with the Comparator tool. Remark 3.5 and the informal claims of Section 7 (unitary evolution, the failure of distributivity, entropy, implementations) are not formalized.

Conventions for Section 7. In Proposition 7.1 a projector is the orthogonal projection `K.starProjection` onto a subspace K of a finite-dimensional complex inner product space, and $P \leq Q$ is inclusion of subspaces; `starProjection_comp_eq_iff` proves that this agrees with $QP = P$, and `starProjection_orthogonal_eq_sub` that $I - P$ projects onto $K^\perp$. In Propositions 7.2 and 7.3 operators are complex matrices, a projector is a Hermitian idempotent matrix, a density operator is a positive semidefinite matrix of trace 1, and the Born probability is $\text{Re Tr}(\rho P)$. In Proposition 7.2 the order $P \leq Q$ is used in its algebraic form $QP = P$, and in Proposition 7.3 $|\uparrow\rangle$ and $|\downarrow\rangle$ are the standard basis vectors of $\mathbb{C}^2$. In part (2) of Proposition 7.3, “ $|\uparrow\rangle$ is the maximum-likelihood state” is proved in the form: the likelihood of $|\uparrow\rangle$ is at least that of every density operator.

Result	Lean declaration (namespace <code>WangLogic</code>)
Definition 3.2	<code>Entails</code>
Definition 3.4	<code>Supported</code>
Proposition 4.1	<code>imp_iff_not_or</code>
Theorem 4.2	<code>deduction_sound</code>
Proposition 4.4	<code>modus_tollens</code>
Proposition 5.1	<code>Supported.of_cons</code>
Theorem 5.2	<code>Supported.mono</code>
Proposition 5.3	<code>falsification</code>
Theorem 5.4	<code>induction_not_deduction</code>
Proposition 5.5	<code>allWhite_supported</code>
Proposition 6.2	<code>m_spec</code>
Proposition 7.1	<code>quantum_deduction_sound</code>
Proposition 7.2(1)	<code>classicalProj_le_iff</code> , <code>classicalProj_not</code>
Proposition 7.2(2)	<code>classical_trace_pos_iff</code>
Proposition 7.3(1)	<code>QSupported.of_concat</code> , <code>likelihood_concat_le</code>
Proposition 7.3(2)	<code>spinState_supported</code> , <code>spinUp_maxLikelihood</code>
Proposition 7.3(3)	<code>spinUp_refuted</code>
Proposition 7.3(4)	<code>likelihood_spinState_phase</code>

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